

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Phase I, Open-Label, Multiple-Dose Study of the Effect of Evobrutinib on the Pharmacokinetics of a Combined Oral Contraceptive in Healthy Female Participants **Short Title/ Acronym:** Effect of Evobrutinib on Pharmacokinetics of a Combined Oral Contraceptive **Protocol Number:** MS200527_0079 **NCT Number:** NCT07215806 **Posting ID:** 766317915096357473 **Trial Status:** COMPLETED **Sponsor/Company Name:** Merck KGaA (EMD Serono) / Merck Healthcare KGaA **Investigational Product:** Evobrutinib (M2951) and Combined oral contraceptive [ethinyl estradiol/ norethisterone (EE/NET)]. Associated trade names for norethisterone include Jinteli, Jevantique, Femhrt, Ortho-Novin, Nor-QD, Aygestin, Combipatch, Norethin 1/50 M, Norinyl 1-50, Myfembree, Tulana, Fyavolv, Norlyroc. **Phase of Development:** Phase 1 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the effect of evobrutinib (M2951) on the pharmacokinetics (PK) of a combined oral contraceptive [Ethinyl estradiol/Norethisterone (EE/NET)] in healthy female participants. **Secondary Objectives:** To evaluate the safety and tolerability of the combined administration of evobrutinib and the oral contraceptive.

2.2 Background & Rationale Understanding the interaction between evobrutinib and oral contraceptives is important because it can help ensure that women using both evobrutinib (a BTK inhibitor under investigation) and this contraceptive have effective and safe contraception. This pharmacokinetic drug-drug interaction (DDI) study aims to establish if the efficacy or safety of the hormonal contraceptive is altered when co-administered with evobrutinib.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Non placebo-controlled Study) **Blinding:** Open-label **Randomization:** N/A (Single-Group Assignment) **Status:** COMPLETED

3.2 Planned Enrollment & Duration Target **SN\$:** 20 healthy female participants **Number of Sites:** Single Clinical Research Unit **Study Duration:** Up to 46 days **Estimated Study Start:** 24-Aug-2022 **Estimated Primary Completion:** 25-Nov-2022

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants are overtly healthy female participants as determined by medical evaluation, including no clinically significant abnormality identified on physical examination or laboratory evaluation and no active clinically significant disorder, condition, infection or disease that would pose a risk to participant safety or interfere with the study evaluation, procedures, or completion. 2. Participants have a body weight within 50.0 and 100.0 kg (inclusive) and body mass index (BMI) within the range 19.0 and 30.0 kg/m² (inclusive). 3. Participants are nonsmokers for at least 6 months preceding Screening. 4. Female participants who are not a Woman of Childbearing Potential (WOCBP). 5. Other protocol defined inclusion criteria could apply.

4.2 Exclusion Criteria 1. Participants with history or presence of clinically relevant respiratory, gastrointestinal, renal, hepatic, hematological, lymphatic, neurological, cardiovascular, psychiatric, musculoskeletal, genitourinary, immunological, dermatological, connective tissue diseases or disorders, as determined by medical evaluation. 2. Participants with diagnosis of hemochromatosis, Wilson's disease, alpha 1 antitrypsin deficiency, or any other chronic liver disease including Gilbert's disease. 3. Participants with prior history of splenectomy or any clinically relevant surgery within 3 months prior to Screening. 4. Participants with history of any malignancy. 5. Participants with history of chronic or recurrent acute infection or any bacterial, viral, parasitic, or fungal infections within 30 days prior to Screening and at any time between Screening and admission, or hospitalization due to infection within 6 months prior to Screening. 6. Participants with history of shingles within 12 months prior to Screening. 7. Administration of live vaccines or live-attenuated virus vaccines within 3 months prior to Screening. Administration of other types of vaccines (e.g., SARS-CoV2 vaccines) is allowed until 4 weeks before admission to CRU, thereafter it is prohibited until the end of the study. Note: In case of clinical symptoms, the participant should be symptom-free for at least 1 week prior to admission to Clinical Research Unit (CRU). 8. Other protocol defined exclusion criteria could apply.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive combined oral contraceptive (COC) [0.03 mg Ethinyl Estradiol (EE) and 0.5 mg Norethisterone (NET)] orally on Day 1 and Day 15 (2 days treatment with COC), in combination with Evobrutinib (M2951) at a dose of 45 mg orally twice daily on Days 4 to 17 (14 days treatment with M2951). **Route of Administration:** Oral **Duration of Treatment:** Total study duration is up to 46

days. Participants are confined to the clinical research unit for an 18-day period (Day -1 to Day 18).

5.2 Concomitant & Prohibited Therapy Permitted: N/A (Restricted due to healthy volunteer PK study constraints). **Prohibited:** Use of any hormonal contraceptives outside of the study protocol prior to and during the evaluation period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures			---
---	---	Screening	Days prior to -1	Informed Consent, Medical Evaluation, BMI/Weight confirmation	Baseline
Day -1	Admission/confinement to the Clinical Research Unit				
Interim	Days 1 to 17	Dosing of COC (Days 1 & 15) and Evobrutinib (Days 4-17); PK blood sampling; Safety monitoring			
End of Study	Day 18	Final PK/Safety assessments, Discharge from Clinical Research Unit			

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definitions: 1. Area Under the Plasma Concentration-time Curve From Time Zero Extrapolated to Infinity (AUC0-inf) of Ethinyl Estradiol and Norethisterone. (AUC0-inf calculated by combining AUC0-t and AUCextra). 2. Maximum Observed Plasma Concentration (Cmax) of Ethinyl Estradiol and Norethisterone. (Obtained directly from the concentration versus time curve). **Measurement Method:** Serial pharmacokinetic (PK) blood sampling and concentration analysis, calculated utilizing non-compartmental analysis rules.

7.2 Secondary & Exploratory Endpoints 1. Number of Participants With Treatment-Emergent Adverse Events (TEAEs). 2. Number of Participants With Treatment-Emergent Adverse Events (TEAEs) by Severity. 3. Change From Baseline in Hematology Parameter: Hemoglobin at Day 18.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during the in-patient stay. TEAEs are defined as events with an onset date or worsening during the on-treatment period. **Serious Adverse Events (SAEs):** Reported immediately if they result in death, are life-threatening, cause persistent/significant disability, or require prolonged inpatient hospitalization. **Data Monitoring Committee (DMC):** Internal safety evaluation managed by the Sponsor's clinical unit.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set and Safety Analysis Set. **Sample Size Justification:** $N=20$ adequately powers the crossover comparison to evaluate the PK curves (AUC and C_{max}) of the contraceptive with and without evobrutinib.

Handling of Missing Data: Missing PK data is handled as per standard non-compartmental analysis rules.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Continuous on-site monitoring by CRU personnel during the 18-day participant confinement period. **Audit Trail:** Secure eCRF entry, sample tracking (blood draw timestamps), and verified data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: MS200527_0079 **Registry Number:** NCT07215806 **Posting ID:** 766317915096357473 **Trial Status:** COMPLETED **Sponsor/Lead Organization:** Merck KGaA (EMD Serono) / Merck **Study Title:** Effect of Evobrutinib on Pharmacokinetics of a Combined Oral Contraceptive **Official Regulatory Title:** A Phase I, Open-Label, Multiple-Dose Study of the Effect of Evobrutinib on the Pharmacokinetics of a Combined Oral Contraceptive in Healthy Female Participants

12. Clinical Framework (Specific)

Primary Condition: Healthy Volunteers / Healthy (Disease Area)

Diagnosis Threshold: Overtly healthy (BMI 19.0 - 30.0 kg/m²)

Medical Methodology: Drug-Drug Interaction (DDI) &

Pharmacokinetic Evaluation (Preferred UMLS Name:

Pharmacokinetics) **Optimization Target:** Assessing clearance, C_{max} , and AUC_{0-inf} of Ethinyl Estradiol/Norethisterone

13. Study Procedures & Methodology

Management Pathway: In-patient confinement at a Clinical Research Unit (CRU) **Education Protocol (HCP):** Standard Phase 1 CRU operational protocol training **Education Protocol (Patient):**

Informed consent detailing strict in-patient CRU rules and contraceptive medication schedule **Quality Audit Mechanism:** Strict timeline adherence for serial PK blood draws and safety evaluations

14. Observation & Follow-Up Schedule

Total Duration: Up to 46 days (with active residency in the CRU from Day -1 to Day 18) **Onsite Visit Cadence:** Continuous confinement from Day -1 to Day 18 **Remote Monitoring Frequency:** N/A (In-patient study) **Checklist Review Interval:** Daily during confinement

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				